

# Rigidity and Quantitative Limits of Phase Averaging in Fabius–Rvachev Quadrature

Signed-filter classification, a rational–irrational dichotomy,  
and sharper superconvergence bounds

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September 28, 2026

## Abstract

We study quadrature obtained by sampling Rvachev’s smooth compactly supported probability density on a translated lattice and then averaging lattice phases. For a rational mesh parameter  $M = a/b$  in lowest terms, we prove that a finite signed phase measure integrates every polynomial of degree at most  $r$ , for *every* overall lattice translation, precisely when it is invariant under rotation by  $1/L$ , where

$$L = b 2^{\max\{0, r - v_2(a)\}}.$$

Consequently, an atomic rule requires at least  $L$  distinct phases. Equality forces the equally weighted regular  $L$ -gon, even when negative weights are allowed. For irrational  $M$ , the only exact phase measure is Haar measure, so no finite phase rule is uniformly exact even on constants. A finite annihilating-polynomial argument strengthens these impossibility statements to explicit positive error bounds for under-budget signed rules.

A second development improves the repository’s odd-dilation estimate for the Fourier product from an inverse first power to an inverse square, and then to a log-Gaussian envelope uniform in the odd part. This gives a uniform relative first-harmonic factorization, a complete integer-mesh phase classification including the odd-mesh boundary case, simple zeros, and quantitative phase-error bounds. Existing phase-classification results in the repository are explicitly credited rather than claimed as new. All conclusions are proved in ordinary mathematics. Exact rational tests, independent Fourier checks, a formalization plan, and further research questions accompany the article. The proposed new refinements and filter theorems have not been independently peer reviewed or checked in Lean.

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# 1 Research target, contributions, and source boundaries

## 1.1 The question that phase averaging raises

Let  $u$  denote the Rvachev up-function normalized to have support  $[-1, 1]$  and integral one. For  $M > 0$  and a phase  $\theta \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$ , consider

$$\mathcal{Q}_{M,\theta}(P) = \frac{1}{M} \sum_{k \in \mathbb{Z}} P\left(\frac{k+\theta}{M}\right) u\left(\frac{k+\theta}{M}\right). \quad (1.1)$$

The sum is finite. On integer meshes the repository proves exactness through degree  $v_2(M)$  at every phase, and one additional degree at certain parity-selected phases. Its prose development also contains stronger phase-zero classifications and multiple-angle estimates [3, 4]. These results naturally raise a different design question: *can several carefully chosen phases, possibly with signed weights, raise the degree of exactness more economically than refining the lattice?*

There are two distinct quantifiers. A single specially chosen phase can cancel an error at one translation. A fixed phase filter that must work *for every overall translation* has to cancel the relevant Fourier coefficients individually. The main results characterize the latter requirement exactly. The answer is rigid: for a rational mesh all admissible phase measures have a prescribed rotational symmetry, and a minimum-size filter is simply a finer lattice in disguise. Moreover, allowing large positive and negative weights cannot make an under-budget rule arbitrarily accurate in the specified uniform moment-error norm.

## 1.2 Principal conclusions

Write  $\mathcal{P}_r$  for the real polynomials of degree at most  $r$ , including zero. For a finite signed Borel measure  $\nu$  on  $\mathbb{T}$  of mass one, define

$$\mathcal{Q}_{M,\theta}^\nu(P) = \int_{\mathbb{T}} \mathcal{Q}_{M,\theta+\varphi}(P) d\nu(\varphi). \quad (1.2)$$

The following are the main statements, with all proofs supplied below.

- 1. Exact signed-filter classification.** If  $M = a/b \in \mathcal{Q}_{>0}$  is reduced and  $L = b2^{\max(0, r-v_2(a))}$ , then (1.2) is exact on  $\mathcal{P}_r$  for every  $\theta$  if and only if  $\nu$  is invariant under  $\varphi \mapsto \varphi + 1/L$  (Theorem 3.2). If  $M$  is irrational, exactness on constants for every  $\theta$  already forces  $\nu$  to be Haar measure (Theorem 3.4).
- 2. Optimal cost and equality cases.** Every exact atomic filter has a number of support points divisible by  $L$ . The minimum is  $L$ , attained only by a rotated uniform  $L$ -gon. Negative weights do not improve this minimum (Theorem 3.5). For an integer mesh, each additional translation-uniform degree requires doubling the minimum phase count.
- 3. A quantitative obstruction below the minimum.** For  $K < L$ , an explicit constant  $c_{M,r,K} > 0$  bounds below the maximum uniform error on the monomials  $1, x, \dots, x^r$  of every  $K$ -phase signed rule (Theorem 4.2). There is an analogous constant-error obstruction for every finite  $K$  on an irrational mesh. Neither result imposes a bound on the weights.
- 4. A sharpened phase theorem.** For  $M = 2^d q$ ,  $q$  odd, the first defect at degree  $p = d + 1$  has a nonvanishing amplitude times its parity-selected sine or cosine, times a positive continuous function lying between  $1 - \eta_p$  and  $1 + \eta_p$ , where

$$\eta_p = (1 - 2^{-p-1})\zeta(p+1) - 1 \leq \pi^2/8 - 1 < 0.234.$$

This gives explicit root conditioning and a unified treatment of  $p = 1$  as well as higher degrees (Theorem 5.4).

### 1.3 What is inherited, and what is proposed here

The classical Fourier product, its integer zero multiplicities, and the probability model are established facts; Arias de Reyna’s paper is a primary reference [1]. Fourier-zero criteria for polynomial reproduction belong to the Strang–Fix tradition; see, for example, the research treatment in [2]. We rederive the particular periodization criterion needed here, rather than assuming that a general reproduction theorem applies without checking its normalization.

The pinned repository’s module `RvachevSuperconvergentSynthesis.lean` explicitly does not assert that its selected phases are the only ones. This is a limitation of that module’s stated surface, *not* evidence that the entire repository lacks a phase-classification proof. Indeed, the canonical comb manuscript contains `thm:phase-zero-set`, including general-odd-part corrected Fabius rules and dyadic Rvachev combs, together with the estimate  $|H(qn/2)| \leq |H(q/2)|/n$  [4]. We therefore do not claim to solve the already-settled dyadic phase conjecture again.

The contribution proposed here is the measure-level classification and its sharp and quantitative cost consequences, together with the inverse-square and uniform log-Gaussian improvements, root estimates, and their synthesis. The stronger comparison uses an additional exactly controlled sinc factor. The proof works down to  $p = 1$ , where the weaker comparison alone does not give the required summable relative sine bound. These are mathematically proved statements in this article, but their worldwide bibliographic novelty is not certified by the targeted literature and repository review. No claim of a solution to a major named open problem is made.

## 2 The kernel and the exact Fourier criterion

### 2.1 Normalization and elementary analytic facts

Our Fourier transform convention is

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx, \quad \text{sinc}(t) = \begin{cases} \sin(\pi t)/(\pi t), & t \neq 0, \\ 1, & t = 0. \end{cases} \quad (2.1)$$

Let  $U_0, U_1, \dots$  be independent random variables uniform on  $[-1/2, 1/2]$ , and set

$$X = \sum_{j \geq 0} 2^{-j} U_j, \quad H(t) = \prod_{j \geq 0} \text{sinc}(t/2^j). \quad (2.2)$$

The series converges absolutely for every outcome and takes values in  $[-1, 1]$ . Its distribution has characteristic function  $H$ . Its density is  $u$ . This agrees with the repository normalization and with the standard up-function [1, 4].

**Lemma 2.1** (Smoothness, product convergence, and zeros). *The distribution of  $X$  has an even nonnegative density  $u \in C_c^\infty(\mathbb{R})$ , with support contained in  $[-1, 1]$  and integral one. The product  $H$  extends to an entire function and converges normally on compact subsets of  $\mathbb{C}$ . Its real zeros are exactly the nonzero integers. At  $m \in \mathbb{Z} \setminus \{0\}$  the order of the zero is*

$$\text{ord}_{t=m} H(t) = 1 + v_2(|m|). \quad (2.3)$$

*In particular  $H(q/2) \neq 0$  for every odd integer  $q$ .*

*Proof.* On each fixed compact set in  $\mathbb{C}$ ,  $\text{sinc}(z/2^j) = 1 + O(4^{-j})$  uniformly. Thus the product is normally convergent. At a point where none of the factors vanishes, the summable perturbation

from one implies that the tail product is nonzero. A factor vanishes precisely when  $z$  is a nonzero integer multiple of  $2^j$ . At a nonzero integer  $m$ , exactly the factors indexed by  $0 \leq j \leq v_2(|m|)$  vanish, and every such zero is simple. This proves (2.3), as well as the absence of any other zeros.

For real  $t \neq 0$  and every positive integer  $N$ , retaining the first  $N$  factors and bounding the rest by one gives

$$|H(t)| \leq \frac{2^{N(N-1)/2}}{(\pi|t|)^N}. \quad (2.4)$$

Consequently  $t^r H(t)$  is integrable for every  $r \geq 0$ . Fourier inversion of the probability distribution therefore gives a smooth density, with all derivatives obtained by differentiating the inverse integral. The support statement follows from the almost-sure bound on  $X$ ; symmetry follows from the symmetric summands. Nonnegativity and normalization follow because this is a probability density. All its derivatives vanish at the endpoints of its support.  $\square$

The value  $u(0) = 1$  can also be seen without a special-function identity: write  $X = U_0 + Y$ , where  $Y$  is supported on  $[-1/2, 1/2]$ . Convolution of the unit-height density of  $U_0$  with the distribution of  $Y$  gives value one at zero; the boundary points of  $Y$  have probability zero. This observation also fixes the central sample in the exact computations below.

## 2.2 Periodized moment defects

For  $r \geq 0$  define

$$\mu_r = \int_{\mathbb{R}} x^r u(x) dx, \quad \mathcal{D}_{M,r}(\theta) = \mathcal{Q}_{M,\theta}(x^r) - \mu_r. \quad (2.5)$$

The finite sum in (1.1) is a smooth periodic function of  $\theta$ : terms entering or leaving the support have every derivative zero. The following formula keeps all scaling and sign conventions explicit.

**Lemma 2.2** (Alias formula). *For every  $M > 0$  and  $r \geq 0$ ,*

$$\mathcal{D}_{M,r}(\theta) = \left(\frac{i}{2\pi}\right)^r \sum_{n \in \mathbb{Z} \setminus \{0\}} H^{(r)}(Mn) e^{2\pi i n \theta}. \quad (2.6)$$

*The series and all of its termwise derivatives in  $\theta$  converge absolutely and uniformly.*

*Proof.* Put  $f_r(x) = x^r u(x)$ . The  $n$ th Fourier coefficient of the periodic sum in (1.1) is, by substituting  $x = (k + \theta)/M$ ,

$$\begin{aligned} \int_0^1 \frac{1}{M} \sum_k f_r((k + \theta)/M) e^{-2\pi i n \theta} d\theta &= \sum_k \int_{k/M}^{(k+1)/M} f_r(x) e^{-2\pi i n M x} dx \\ &= \widehat{f_r}(Mn). \end{aligned}$$

There is no extra power of  $M$  in this coefficient. Differentiating  $H$  under its integral gives  $\widehat{f_r}(t) = (i/(2\pi))^r H^{(r)}(t)$ . Since  $f_r$  is smooth and compactly supported, repeated integration by parts shows that  $\widehat{f_r}$  decreases faster than any fixed inverse power. This proves all convergence assertions and identifies the Fourier series. Its zero coefficient is  $\mu_r$ , so subtracting it yields (2.6).  $\square$

**Corollary 2.3** (Unfiltered, translation-uniform exactness). *An unfiltered rule at mesh  $M > 0$  integrates  $\mathcal{P}_r$  for every phase if and only if  $M$  is a positive integer and  $r \leq v_2(M)$ .*

*Proof.* Exactness on constants requires  $H(M) = 0$ , so  $M$  must be an integer. For an integer  $M$ , all derivatives of orders at most  $r$  vanish at every  $Mn$ ,  $n \neq 0$ , precisely when  $r < 1 + v_2(M)$ . The necessity of this inequality is witnessed already by  $n = 1$  in (2.6).  $\square$

### 3 Classification of all exact signed phase filters

#### 3.1 Multipliers of a finite signed measure

Throughout this section a finite signed measure means a real, countably additive Borel measure with finite total variation. For such a measure  $\nu$  on  $\mathbb{T}$  of total mass one, set

$$a_n = \int_{\mathbb{T}} e^{2\pi i n \varphi} d\nu(\varphi), \quad n \in \mathbb{Z}. \quad (3.1)$$

Thus  $a_0 = 1$ ,  $a_{-n} = \overline{a_n}$ , and  $|a_n| \leq \|\nu\|_{TV}$ . We say  $\nu$  is *uniformly exact through  $r$  at mesh  $M$*  when

$$\mathcal{Q}_{M,\theta}^\nu(P) = \int_{\mathbb{R}} P(x)u(x) dx \quad \text{for all } P \in \mathcal{P}_r \text{ and all } \theta \in \mathbb{T}. \quad (3.2)$$

The same  $\nu$  must work for all polynomials and all translations.

**Lemma 3.1** (Filtered Fourier test). *Condition (3.2) is equivalent to*

$$a_n H^{(j)}(Mn) = 0 \quad (n \neq 0, 0 \leq j \leq r). \quad (3.3)$$

*Equivalently,  $a_n$  may be nonzero only at indices for which  $Mn$  is a nonzero integer and  $v_2(|Mn|) \geq r$ .*

*Proof.* Integrate the uniformly absolutely convergent series (2.6) against  $\nu$ . Finite total variation justifies the interchange and gives

$$\mathcal{D}_{M,j}^\nu(\theta) := \mathcal{Q}_{M,\theta}^\nu(x^j) - \mu_j = \left(\frac{i}{2\pi}\right)^j \sum_{n \neq 0} a_n H^{(j)}(Mn) e^{2\pi i n \theta}. \quad (3.4)$$

A continuous periodic function vanishes identically if and only if all its Fourier coefficients vanish. Exactness on the monomial basis is therefore equivalent to (3.3). If  $Mn$  is not an integer,  $H(Mn) \neq 0$  already forces  $a_n = 0$ . At a nonzero integer of zero order  $k$ , all derivatives through  $r$  vanish exactly when  $k > r$ ; otherwise the derivative of order  $k$  is nonzero. Apply Lemma 2.1.  $\square$

#### 3.2 Rational meshes: a cyclic symmetry is necessary and sufficient

**Theorem 3.2** (Rational-mesh rigidity). *Let  $M = a/b > 0$ , where  $a, b \in \mathbb{N}_{>0}$  and  $\gcd(a, b) = 1$ . Let  $r \geq 0$  and put*

$$d = v_2(a), \quad L = b 2^{\max\{0, r-d\}}. \quad (3.5)$$

*For a finite signed Borel measure  $\nu$  on  $\mathbb{T}$  of mass one, the following are equivalent:*

- (i)  $\nu$  is uniformly exact through degree  $r$  at mesh  $M$ ;
- (ii)  $a_n = 0$  for every  $n \notin L\mathbb{Z}$ ;
- (iii)  $\nu$  is invariant under rotation by  $1/L$ .

*For  $L = 1$ , the rotation is the identity and (ii) imposes no restriction.*

*Proof.* First identify the permitted frequencies in Lemma 3.1. The condition  $Mn \in \mathbb{Z}$  is equivalent to  $b \mid n$  because  $a/b$  is reduced. Writing  $n = bk$ , the remaining condition is

$$v_2(|ak|) = d + v_2(|k|) \geq r.$$

This says exactly that  $2^{\max(0, r-d)}$  divides  $k$ , or  $L$  divides  $n$ . This proves (i)  $\Leftrightarrow$  (ii).

Let  $R_h\nu$  be the pushforward of  $\nu$  under  $\varphi \mapsto \varphi + h$ . Its multiplier is  $e^{2\pi i n h} a_n$ . Invariance under  $h = 1/L$  therefore implies that  $a_n = 0$  whenever  $L \nmid n$ . Conversely, if (ii) holds,  $R_{1/L}\nu$  and  $\nu$  have all the same Fourier coefficients. Trigonometric polynomials are uniformly dense in continuous functions on  $\mathbb{T}$ . Integrating their uniform approximations against these finite-variation measures shows that the measures agree on every continuous function, and hence are equal. This proves (ii)  $\Leftrightarrow$  (iii).  $\square$

The theorem classifies singular continuous measures as well as atomic and absolutely continuous ones. If  $\nu$  has a density, that density must be  $1/L$ -periodic almost everywhere. In the signed setting there is no positivity premise hidden in the Fourier argument.

**Corollary 3.3** (Symmetrization description). *At a rational mesh with the integer  $L$  in (3.5), every measure of the form*

$$\nu = \frac{1}{L} \sum_{j=0}^{L-1} R_{j/L} \rho, \quad \rho(\mathbb{T}) = 1, \quad (3.6)$$

*is uniformly exact through  $r$ . Conversely every exact measure has this form, for example by taking  $\rho = \nu$ .*

*Proof.* The averaging operator in (3.6) has as its fixed points precisely the  $1/L$ -rotation-invariant measures.  $\square$

### 3.3 Irrational meshes: continuous averaging is unavoidable

**Theorem 3.4** (Rational–irrational dichotomy). *Let  $M > 0$  be irrational. A finite signed phase measure of mass one is uniformly exact on constants if and only if it is normalized Haar measure  $m_{\mathbb{T}}$ . Haar averaging, in fact, integrates  $P(x)u(x)$  exactly for every polynomial  $P$ . No finite atomic phase measure is uniformly exact on constants at such a mesh.*

*Proof.* For every nonzero integer  $n$ ,  $Mn$  is not an integer and so  $H(Mn) \neq 0$ . Lemma 3.1, even with  $r = 0$ , gives  $a_n = 0$  for all  $n \neq 0$ . Together with  $a_0 = 1$ , these are the Fourier coefficients of normalized Haar measure. Fourier uniqueness, as justified in the proof of Theorem 3.2, gives  $\nu = m_{\mathbb{T}}$ .

Conversely, directly unfolding the integral yields, for every integrable function  $f$ ,

$$\int_0^1 \frac{1}{M} \sum_{k \in \mathbb{Z}} f((k + \theta + \varphi)/M) d\varphi = \int_{\mathbb{R}} f(x) dx. \quad (3.7)$$

For a signed integrable function this follows by applying the positive identity to its positive and negative parts. Take  $f = Pu$ . Haar measure is nonatomic and cannot equal a finitely supported measure of mass one.  $\square$

### 3.4 Minimum support, signed weights, and equality cases

**Theorem 3.5** (Optimal atomic phase cost). *Let  $M = a/b$  and  $L$  be as in Theorem 3.2. Suppose*

$$\nu = \sum_{j=1}^K w_j \delta_{\varphi_j}, \quad w_j \in \mathbb{R} \setminus \{0\}, \quad \sum_j w_j = 1, \quad (3.8)$$

where the phases are distinct modulo one. If  $\nu$  is uniformly exact through  $r$ , then  $L$  divides  $K$ , and in particular  $K \geq L$ . All such atomic measures are unions of complete  $L$ -cycles, with equal weights within each cycle. When  $K = L$  there is an  $\alpha \in \mathbb{T}$  such that

$$\nu = \frac{1}{L} \sum_{j=0}^{L-1} \delta_{\alpha+j/L}. \quad (3.9)$$

Thus every minimum-support exact signed filter is positive.

*Proof.* By Theorem 3.2, rotation by  $1/L$  leaves  $\nu$  unchanged. The orbit of every point under this rotation has exactly  $L$  points. If an atom has nonzero mass  $w$ , all points in its orbit are atoms of mass  $w$ . Distinct orbits are disjoint, so the support is a disjoint union of  $L$ -point orbits and  $L \mid K$ . If  $K = L$ , there is just one orbit, and its common mass is  $1/L$  because the total mass is one. Conversely any union of such equally weighted orbits with total mass one is invariant and hence exact.  $\square$

**Proposition 3.6** (The optimum is a finer lattice). *The measure in (3.9) satisfies*

$$\mathcal{Q}_{M,\theta}^\nu(P) = \mathcal{Q}_{LM, L(\theta+\alpha)}(P). \quad (3.10)$$

*Proof.* The left side is

$$\frac{1}{LM} \sum_{j=0}^{L-1} \sum_{k \in \mathbb{Z}} (Pu) \left( \frac{Lk + j + L(\theta + \alpha)}{LM} \right).$$

Each integer has a unique representation  $Lk + j$ ,  $0 \leq j < L$ . Reindexing gives (3.10).  $\square$

For an integer mesh  $M = 2^d q$  with  $q$  odd, a gain of  $s \geq 0$  translation-uniform degrees, from  $d$  to  $d + s$ , costs exactly  $2^s$  phases. More generally, with a budget of at most  $K$  phases at the rational mesh  $a/b$ , no uniform constant rule exists if  $K < b$ . If  $K \geq b$ , the largest attainable uniform degree is

$$r_{\max} = v_2(a) + \lfloor \log_2(K/b) \rfloor. \quad (3.11)$$

Indeed, the requirement  $b2^{\max(0, r-d)} \leq K$  is necessary by Theorem 3.5 and sufficient by (3.9).

*Remark 3.7* (What this lower bound does not say). The bound is on distinct *phase shifts* of the fixed kernel rule, with the weights fixed independently of the overall translation. It is not a lower bound for arbitrary Gaussian quadrature nodes, nor for a rule designed at just one phase, nor for weights allowed to vary with  $\theta$  or with the polynomial. These changes alter the problem's quantifiers and invalidate the coefficientwise necessity argument.

## 4 A quantitative obstruction for under-budget signed filters

The preceding support count concerns exact equality. A stronger statement is possible: for fixed  $M, r, K$  below the threshold, the entire class of  $K$ -phase signed rules has a strictly positive lower bound on uniform moment error. The weights need not lie in a compact or bounded set.

### 4.1 A finite annihilator certificate

**Lemma 4.1** (A nonzero early multiplier). *Let  $\nu$  be as in (3.8). Then*

$$\max_{1 \leq n \leq K} |a_n| \geq \frac{1}{2^K - 1}. \quad (4.1)$$

More precisely, if  $z_j = e^{2\pi i \varphi_j}$  and

$$A(z) = \prod_{j=1}^K (z - z_j) = \sum_{n=0}^K c_n z^n, \quad (4.2)$$

then  $|c_0| = 1$ ,  $\sum_{n=1}^K |c_n| \leq 2^K - 1$ , and

$$\sum_{n=1}^K c_n a_n = -c_0. \quad (4.3)$$

*Proof.* The polynomial  $A$  vanishes at every  $z_j$ . Multiply  $A(z_j) = 0$  by  $w_j$  and sum. Since  $a_0 = 1$ , this gives (4.3). The roots have modulus one, so  $|c_0| = 1$  and the elementary symmetric coefficient bound is  $|c_n| \leq \binom{K}{n}$ . Hence

$$1 \leq \sum_{n=1}^K |c_n| |a_n| \leq (2^K - 1) \max_{1 \leq n \leq K} |a_n|.$$

No step uses positivity of the weights. In fact the conclusion also holds for complex weights of total mass one.  $\square$

## 4.2 Explicit uniform error bounds

Define the dimensionless moment-error norm

$$E_{M,r}(\nu) = \max_{0 \leq j \leq r} \left\| \mathcal{D}_{M,j}^\nu \right\|_{L^\infty(\mathbb{T})}. \quad (4.4)$$

The coordinate  $x$  is already normalized to the fixed support  $[-1, 1]$ .

**Theorem 4.2** (Quantitative under-budget obstruction). *Let  $M = a/b$  be rational, let  $L$  be given by (3.5), and let  $1 \leq K < L$ . For  $1 \leq n \leq K$  define*

$$j_n = \begin{cases} 0, & Mn \notin \mathbb{Z}, \\ 1 + v_2(Mn), & Mn \in \mathbb{Z}, \end{cases} \quad B_n = (2\pi)^{-j_n} |H^{(j_n)}(Mn)|. \quad (4.5)$$

Then  $0 \leq j_n \leq r$  and  $B_n > 0$ . Every signed measure of mass one supported on at most  $K$  phases satisfies

$$E_{M,r}(\nu) \geq \frac{\min_{1 \leq n \leq K} B_n}{2^K - 1} > 0. \quad (4.6)$$

For an irrational  $M > 0$  and every  $K \geq 1$ , the corresponding conclusion already holds on constants:

$$\left\| \mathcal{D}_{M,0}^\nu \right\|_\infty \geq \frac{\min_{1 \leq n \leq K} |H(Mn)|}{2^K - 1} > 0. \quad (4.7)$$

*Proof.* In the rational case, none of the indices  $1, \dots, K$  lies in  $L\mathbb{Z}$ . The frequency calculation in Theorem 3.2 shows that if  $Mn$  is an integer, its zero order is at most  $r$ ; otherwise order zero is a nonvanishing witness. Thus (4.5) is defined with  $j_n \leq r$  and  $B_n > 0$ .

If the support has fewer than  $K$  points, multiply its annihilating polynomial by additional factors with unit-modulus roots to reach degree  $K$ . The coefficient proof of Lemma 4.1 remains valid. The  $n$ th Fourier coefficient of the  $j_n$ th filtered defect has modulus  $B_n |a_n|$  by (3.4). The modulus of a Fourier coefficient of a continuous periodic function is at most its uniform norm, so

$$E_{M,r}(\nu) \geq B_n |a_n| \quad (1 \leq n \leq K).$$

Combine this with Lemma 4.1 to obtain (4.6). More specifically, the actual nodes give

$$E_{M,r}(\nu) \geq \left( \sum_{n=1}^K \frac{|c_n|}{B_n} \right)^{-1}, \quad (4.8)$$

by inserting  $|a_n| \leq E_{M,r}(\nu)/B_n$  into (4.3). Formula (4.6) is a node-independent consequence.

For irrational  $M$ , each  $H(Mn)$  is nonzero. Use  $j_n = 0$  for every  $n$  and repeat the same argument. Positivity of the minimum follows from taking a minimum over a nonempty finite set of strictly positive numbers, not from a global lower bound for  $H$ .  $\square$

*Remark 4.3* (Explicit does not mean numerically large). The constants can be very small, because  $H$  decays rapidly and because  $2^K - 1$  grows. The conclusion is a rigorous positive obstruction for *fixed* parameters, not a claim of a practical lower error floor uniform in  $M$  or  $K$ . Near a rational mesh, the irrational constant can tend to zero. That behavior is consistent with the exact rational–irrational dichotomy and prevents a false claim of a discontinuous positive universal error threshold.

**Corollary 4.4** (Integer-mesh certificate without high derivatives). *Suppose  $M = 2^d q$  with  $q$  odd,  $r = d + s$ , and  $K < 2^s$ . For  $1 \leq n \leq K$ , write  $n = 2^e \ell$  with  $\ell$  odd and put  $j_n = d + e + 1$ . Then the constants in (4.5) are*

$$B_n = \frac{j_n! |H(q\ell/2)|}{(2\pi Mn)^{j_n}}. \quad (4.9)$$

*Proof.* At any positive integer  $t = 2^v q_0$  with  $q_0$  odd, the leading-zero calculation in Lemma 5.3 below gives  $H^{(v+1)}(t) = -(v+1)!H(q_0/2)/t^{v+1}$ . Apply it to  $t = Mn = 2^{d+e}q\ell$ .  $\square$

These certificates are finite in a useful sense: only the first  $K$  phase multipliers and the corresponding nonzero kernel coefficients are required. One need not search all Fourier frequencies to find a witness that an under-budget rule fails.

## 5 Sharper odd-dilation bounds and pointwise superconvergence

### 5.1 An inverse-square comparison

The repository’s multiple-angle argument isolates one exactly controlled factor of the product and obtains an inverse first power [4]. Two factors can be controlled exactly, uniformly in the odd part of the mesh.

**Lemma 5.1** (Odd-dilation inverse-square bound). *For all positive odd integers  $q, n$ ,*

$$0 < \left| \frac{H(qn/2)}{H(q/2)} \right| \leq \frac{1}{n^2}. \quad (5.1)$$

*Proof.* For every positive integer  $n$  and real  $x$ ,  $|\sin(nx)| \leq n|\sin x|$ . For example, this follows from  $e^{inx} - e^{-inx} = (e^{ix} - e^{-ix}) \sum_{j=0}^{n-1} e^{i(n-1-2j)x}$  and the triangle inequality. Consequently  $|\operatorname{sinc}(ny)| \leq |\operatorname{sinc}(y)|$  for every real  $y$ .

Compare the factors of  $H(qn/2)$  and  $H(q/2)$ . At product index zero,  $q$  and  $qn$  are odd, so the absolute sine numerators are both one. The factor ratio is exactly  $1/n$ . At index one, the arguments of the sine are  $\pi q/4$  and  $\pi qn/4$ ; for every odd integer  $h$ ,  $|\sin(\pi h/4)| = 1/\sqrt{2}$ . This second factor ratio is also exactly  $1/n$ . Every remaining factor ratio is at most one by the multiple-angle inequality. No denominator factor vanishes. The comparison holds for every finite product with at least two factors, and passing to the limit gives the upper bound. Both full products are nonzero by Lemma 2.1, which gives the strict lower inequality.  $\square$

The improvement is especially useful when a derivative of a trigonometric harmonic costs one power of its frequency: the inverse square still leaves a summable majorant at the lowest threshold degree. In fact the arithmetic separation from zeros at every dyadic scale gives a stronger uniform result.

**Proposition 5.2** (A uniform multiscale odd-dilation envelope). *For positive odd  $q$  and odd  $n \geq 3$ , put  $m = \lfloor \log_2 n \rfloor$ . Then*

$$\left| \frac{H(qn/2)}{H(q/2)} \right| \leq \frac{2^{m(m+1)/2-1}}{n^{m+1}} \leq \frac{1}{2\sqrt{n}} \exp\left(-\frac{(\log n)^2}{2 \log 2}\right). \quad (5.2)$$

*In particular the relative odd-dilation ratio decreases faster than every fixed inverse power of  $n$ , uniformly over all positive odd  $q$ .*

*Proof.* At index  $j \geq 2$ , oddness of  $q$  gives

$$|\sin(\pi q/2^{j+1})| \geq \sin(\pi/2^{j+1}) \geq 2^{-j}.$$

The first inequality follows by reducing the odd numerator modulo  $2^{j+1}$  and measuring its distance from the nearest multiple of that integer. The second is  $\sin x \geq 2x/\pi$  on  $[0, \pi/2]$ . Thus the absolute ratio of the  $j$ th factors is at most  $2^j/n$ , as well as at most one by the multiple-angle inequality. Retain the exact  $n^{-2}$  from the first two factors and the bounds  $2^j/n$  for  $2 \leq j \leq m$ . All later ratios are at most one. Their product is

$$n^{-2} \prod_{j=2}^m \frac{2^j}{n} = 2^{m(m+1)/2-1} n^{-m-1},$$

with the product empty when  $m = 1$ . For the second inequality write  $\log_2 n = m + t$ ,  $0 \leq t < 1$ . The logarithm of this upper bound is

$$-\frac{(\log n)^2}{2 \log 2} - \frac{1}{2} \log n - \log 2 + \frac{\log 2}{2} (t^2 - t),$$

and the final term is nonpositive. The superalgebraic conclusion follows because a negative quadratic in  $\log n$  dominates every fixed negative linear term in  $\log n$ .  $\square$

This is a relative estimate: it remains valid when  $H(q/2)$  itself is extremely small. An ordinary decay estimate for  $H(t)$  at large  $t$  would not by itself provide this uniformity after dividing by  $H(q/2)$ . For the root theorem below we retain the simpler inverse-square bound, which makes the constants transparent; (5.2) can further reduce them.

## 5.2 Leading jets at all integer zeros

**Lemma 5.3** (The first nonzero derivative). *Let  $t_0 = 2^v q_0$  be a nonzero integer,  $q_0$  odd, and put  $k = v + 1$ . Then*

$$H^{(k)}(t_0) = -\frac{k!}{t_0^k} H(q_0/2). \quad (5.3)$$

*In particular, if  $M = 2^d q$  with  $q > 0$  odd and  $p = d + 1$ , then for odd  $n \in \mathbb{Z}$ ,*

$$H^{(p)}(Mn) = -\frac{p!}{(Mn)^p} H(qn/2), \quad (5.4)$$

*whereas  $H^{(p)}(Mn) = 0$  for nonzero even  $n$ .*

*Proof.* Split the product into its  $k$  vanishing factors and the tail:

$$H(t) = \prod_{j=0}^v \operatorname{sinc}(t/2^j) H(t/2^{v+1}).$$

At  $t = t_0$ , the linear coefficient of the  $j$ th factor is  $(-1)^{t_0/2^j}/t_0$ . The product of their signs is

$$(-1)^{\sum_{j=0}^v t_0/2^j} = (-1)^{q_0(2^{v+1}-1)} = -1.$$

Thus the coefficient of  $(t - t_0)^k$  is  $-t_0^{-k} H(q_0/2)$ , proving (5.3). For even  $n \neq 0$ , the zero order at  $Mn$  is at least  $d + 2 = p + 1$ , so its  $p$ th derivative vanishes.  $\square$

This leading-jet identity is already represented in the repository's `CombFirstDefect.lean` development [4]. Its derivation is included to make the signs and constants of the new bounds auditable.

### 5.3 A positive relative factor and complete zero sets

Fix  $M = 2^d q$  with  $d \geq 0$  and  $q > 0$  odd. Write

$$p = d + 1, \quad \tau_p(t) = \begin{cases} \sin t, & p \text{ odd}, \\ \cos t, & p \text{ even}, \end{cases} \quad \epsilon_p = (-1)^{\lfloor (p-1)/2 \rfloor}, \quad (5.5)$$

and define the nonzero amplitude

$$A_{M,p} = \frac{2p! \epsilon_p H(q/2)}{(2\pi M)^p}. \quad (5.6)$$

Pairing positive and negative odd aliases in (2.6) gives

$$\mathcal{D}_{M,p}(\theta) = A_{M,p} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{H(qn/2)}{H(q/2)n^p} \tau_p(2\pi n\theta). \quad (5.7)$$

For instance, the sign factors for  $p = 1, 2, 3, 4$  are respectively  $+, +, -, -$ ; this is a useful independent normalization check.

**Theorem 5.4** (Relative factorization and all integer-mesh phase zeros). *With the notation above, there exists a continuous one-periodic real function  $G_{p,q}$  satisfying*

$$\mathcal{D}_{M,p}(\theta) = A_{M,p} \tau_p(2\pi\theta) G_{p,q}(\theta), \quad (5.8)$$

$$1 - \eta_p \leq G_{p,q}(\theta) \leq 1 + \eta_p, \quad \eta_p = \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-p-1} = (1 - 2^{-p-1})\zeta(p+1) - 1. \quad (5.9)$$

*In particular  $G_{p,q}$  is strictly positive, since  $\eta_p \leq \pi^2/8 - 1 < 1$ . The rule  $\mathcal{Q}_{M,\theta}$  integrates  $\mathcal{P}_{d+1}$  exactly if and only if, modulo one,*

$$\theta \in Z_p := \begin{cases} \{0, 1/2\}, & d \text{ even}, \\ \{1/4, 3/4\}, & d \text{ odd}. \end{cases} \quad (5.10)$$

*At any other phase its degree of precision is exactly  $d$ .*

*Proof.* For odd  $n$  the quotients

$$R_{p,n}(t) = \frac{\tau_p(nt)}{\tau_p(t)}$$

extend continuously across the zeros of the denominator. In the sine case, the quotient is the polynomial  $U_{n-1}(\cos t)$ , where  $U$  is the Chebyshev polynomial of the second kind; equivalently the elementary multiple-angle identity proves the same divisibility. In the cosine case,  $\cos(nt)$  is an odd polynomial in  $\cos t$  because  $n$  is odd, so division by  $\cos t$  also leaves a polynomial. The sine inequality in the proof of Lemma 5.1 gives  $|R_{p,n}(t)| \leq n$ . For cosines use the same inequality after shifting  $t$  by  $\pi/2$ ; oddness of  $n$  changes only a sign.

Define

$$G_{p,q}(\theta) = 1 + \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{H(qn/2)}{H(q/2)n^p} R_{p,n}(2\pi\theta). \quad (5.11)$$

By Lemma 5.1, the absolute value of each summand is at most  $n^{-p-1}$ . Since  $p \geq 1$ , the series is uniformly convergent. It defines a continuous real periodic function and obeys (5.9). Multiplying it by  $\tau_p(2\pi\theta)$  gives (5.7), including at the zeros by continuity. This proves (5.8).

All degrees at most  $d$  are exact at all phases by Corollary 2.3. The only extra monomial to check for  $\mathcal{P}_{d+1}$  is  $x^p$ . Its defect vanishes exactly when  $\tau_p(2\pi\theta) = 0$ , because both  $A_{M,p}$  and  $G_{p,q}$  are nonzero. These are exactly the phases in (5.10). Outside them,  $x^{d+1}$  witnesses failure, proving the final assertion.  $\square$

*Remark 5.5* (The lowest-degree endpoint matters). The theorem includes every odd mesh  $M = q$ , where  $d = 0$  and  $p = 1$ . A comparison of size  $n^{-1}$  alone would bound the relative sine tail by a divergent harmonic-type sum in this boundary case. The second exact sinc ratio makes the relevant majorant  $n^{-2}$  and settles the boundary case by the same argument as every other integer mesh. For higher degrees, the result also sharpens the quantitative constants available from the weaker comparison. The statement does not assert that the selected phases fail at the very next degree.

## 5.4 Simple zeros, phase conditioning, and arithmetic signs

**Corollary 5.6** (Simple zeros with uniform relative derivative bounds). *Every zero  $\theta_* \in Z_p$  of  $\mathcal{D}_{M,p}$  is simple, and*

$$2\pi|A_{M,p}|(1 - \eta_p) \leq |\mathcal{D}'_{M,p}(\theta_*)| \leq 2\pi|A_{M,p}|(1 + \eta_p). \quad (5.12)$$

*Proof.* The defect is smooth by Lemma 2.2. Divide (5.8) by  $\theta - \theta_*$  and pass to the limit. Continuity of  $G_{p,q}$  suffices, and  $|\frac{d}{d\theta}\tau_p(2\pi\theta_*)| = 2\pi$ . Now use (5.9). No unproved differentiability of the quotient series at a root is being assumed.  $\square$

**Corollary 5.7** (Global distance-to-phase bounds). *Let  $\rho = \text{dist}_{\mathbb{T}}(\theta, Z_p)$ , so  $0 \leq \rho \leq 1/4$ . Then*

$$4|A_{M,p}|(1 - \eta_p)\rho \leq |\mathcal{D}_{M,p}(\theta)| \leq 2\pi|A_{M,p}|(1 + \eta_p)\rho. \quad (5.13)$$

*Consequently, an observed bound  $|\mathcal{D}_{M,p}(\theta)| \leq \varepsilon$  implies*

$$\text{dist}_{\mathbb{T}}(\theta, Z_p) \leq \frac{\varepsilon}{4|A_{M,p}|(1 - \eta_p)}. \quad (5.14)$$

*Proof.* For the nearest zero,  $|\tau_p(2\pi\theta)| = \sin(2\pi\rho)$ . On  $[0, \pi/2]$ , concavity gives  $\sin x \geq 2x/\pi$ , while  $\sin x \leq x$ . Combine these with (5.8).  $\square$

The dependence on the amplitude is essential. The roots are well conditioned after dividing by  $A_{M,p}$ , but the absolute defect can be extremely small for large odd  $q$ . An unnormalized absolute error test alone may then give a weak certificate for the actual phase.

**Proposition 5.8** (Binary sign rule). *Let  $s_2(q)$  be the number of ones in the binary expansion of a positive odd integer  $q$ . Then*

$$\operatorname{sgn} H(q/2) = (-1)^{s_2(q)-1}. \quad (5.15)$$

For  $\theta \notin Z_p$ ,

$$\operatorname{sgn} \mathcal{D}_{M,p}(\theta) = (-1)^{\lfloor (p-1)/2 \rfloor + s_2(q)-1} \operatorname{sgn} \tau_p(2\pi\theta). \quad (5.16)$$

*Proof.* The sign of the  $j$ th sinc factor at  $q/2$  is  $(-1)^{\lfloor q/2^{j+1} \rfloor}$ . All but finitely many are positive. If  $q = \sum_{h \geq 0} b_h 2^h$ , then

$$\sum_{k \geq 1} \lfloor q/2^k \rfloor = \sum_{h \geq 1} b_h (2^h - 1) = q - s_2(q).$$

Since  $q$  is odd,  $q - s_2(q)$  has the same parity as  $s_2(q) - 1$ . This proves (5.15). Equation (5.16) follows from the positive factor in (5.8).  $\square$

The binary sign phenomenon itself is part of the existing repository analysis [4]; the point here is its combination with the sharper global magnitude and root estimates.

## 6 Consequences for polynomial synthesis and concrete examples

### 6.1 Triangular moment deconvolution

The repository transports quadrature identities into polynomial synthesis using inverse moment operators [3]. The filter theorems respect this transport exactly; they are not restricted to monomials as a preferred application.

Define the operator on polynomials

$$(TP)(y) = \int_{\mathbb{R}} P(y-x)u(x) \, dx = \sum_{j \geq 0} \frac{(-1)^j \mu_j}{j!} P^{(j)}(y), \quad (6.1)$$

where the sum is finite. Since  $\mu_0 = 1$ ,  $T$  preserves the leading coefficient and is an invertible triangular operator on every  $\mathcal{P}_r$ . Indeed  $T = I + N$  with  $N$  degree-lowering, so on  $\mathcal{P}_r$ ,  $T^{-1} = \sum_{k=0}^r (-N)^k$ .

**Proposition 6.1** (Exact transfer, including necessity). *Fix  $M > 0$ , a phase measure  $\nu$ , and  $r \geq 0$ . The uniform quadrature identity (3.2) is equivalent to*

$$P(y) = \int_{\mathbb{T}} \frac{1}{M} \sum_{k \in \mathbb{Z}} (T^{-1}P) \left( y - \frac{k + \theta + \varphi}{M} \right) u \left( \frac{k + \theta + \varphi}{M} \right) \, d\nu(\varphi) \quad (6.2)$$

for all  $P \in \mathcal{P}_r$ , all  $y \in \mathbb{R}$ , and all  $\theta \in \mathbb{T}$ .

*Proof.* If quadrature is exact, apply it to  $R_y(x) = (T^{-1}P)(y-x)$ , whose degree is at most  $r$ . Its integral against  $u$  is  $T(T^{-1}P)(y) = P(y)$ . Conversely fix any  $y$ . As  $P$  ranges through  $\mathcal{P}_r$ , invertibility of  $T^{-1}$  and of the substitution  $z \mapsto y-z$  implies that  $R_y$  ranges through all of  $\mathcal{P}_r$ . Thus (6.2) is precisely quadrature exactness for every such polynomial.  $\square$

Consequently the same cyclic invariance, minimum number of phases, and irrational obstruction hold for this deconvolved synthesis scheme. The selected-phase criterion in Theorem 5.4 likewise transfers to a single phase at the first threshold degree.

## 6.2 Phase-cost examples

**Example 6.2** (An odd denominator). Take  $M = 3/5$ . Then  $v_2(3) = 0$ . Uniform exactness through degree two requires  $L = 5 \cdot 2^2 = 20$  phases, even with signed weights. At the minimum, the phases are  $\alpha + j/20$  with weights  $1/20$ , and the resulting rule is the mesh-12 rule with phase  $20(\theta + \alpha)$ . Since  $v_2(12) = 2$ , the degree matches exactly. A rule with at most 19 phases has a positive uniform moment-error lower bound supplied by Theorem 4.2.

**Example 6.3** (Integer mesh versus special phase). For  $M = 12$ ,  $d = 2$  and  $p = 3$ . Every phase integrates quadratics. Exactly phases 0 and  $1/2$  modulo one integrate cubics as well. To integrate cubics for every overall translation using a fixed filter requires two antipodal phases of equal weight. Uniform degree five requires eight phases; at the minimum this is the mesh-96 rule. The phase count is about translation-uniform exactness, not about the two exceptional cubic phases.

**Example 6.4** (A concrete positive obstruction). Take  $M = 1$ , target degree  $r = 1$ , and  $K = 1$ . Here  $L = 2$ ,  $j_1 = 1$ , and

$$E_{1,1}(\nu) \geq \frac{|H(1/2)|}{2\pi}.$$

Since a one-point signed measure of mass one has weight one, this is a lower bound valid for every translation of that single phase. At a *fixed* phase, however, the linear moment is exact at 0 and  $1/2$ . This example explicitly separates the quantifiers responsible for the obstruction.

**Example 6.5** (Irrational scale). For  $M = \sqrt{2}$  and any  $K$ -phase signed filter,

$$\sup_{\theta} |\mathcal{Q}_{\sqrt{2},\theta}^\nu(1) - 1| \geq \frac{\min_{1 \leq n \leq K} |H(n\sqrt{2})|}{2^K - 1} > 0.$$

No finite filter can produce a partition of unity at all translations. Continuous uniform phase averaging does so exactly by (3.7).

## 7 Exact evaluation and reproducible verification

### 7.1 A moment recurrence

The verification program does not use a truncated infinite product to certify rational quadrature values. It evaluates dyadic samples exactly using finite convolution cells and rational moments.

**Lemma 7.1** (Rational moments). *The moments satisfy  $\mu_0 = 1$ ,  $\mu_{2n+1} = 0$ , and for  $n \geq 1$ ,*

$$\mu_{2n} = \frac{1}{2^{2n} - 1} \sum_{j=1}^n \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}. \quad (7.1)$$

*In particular all moments are rational, with*

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}.$$

*Proof.* The random series has the distributional identity  $X \stackrel{d}{=} (V + X')/2$ , where  $V$  is uniform on  $[-1, 1]$  and independent of the copy  $X'$  of  $X$ . Expand the  $2n$ th power. Odd moments vanish by symmetry and  $\mathbb{E}V^{2j} = 1/(2j+1)$ . Moving the term involving  $\mu_{2n}$  to the left gives (7.1).  $\square$

## 7.2 A finite exact dyadic evaluator

For a nonnegative integer  $m$ , write  $\sigma(m) = (-1)^{s_2(m)}$ . For  $N \geq 1$  and real  $x$ , put

$$a_m(x) = x + 1 - 2^{-N} - m2^{-(N-1)}, \quad 0 \leq m < 2^N. \quad (7.2)$$

**Proposition 7.2** (Exact dyadic evaluation formula). *Suppose  $x$  is dyadic with  $|x| < 1$ , and write its reduced denominator as  $2^d$ . Set  $N = d + 1$ . Then*

$$u(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{\substack{0 \leq m < 2^N \\ a_m(x) > 0}} \sigma(m) \sum_{\substack{0 \leq j \leq N-1 \\ j \text{ even}}} \binom{N-1}{j} a_m(x)^{N-1-j} 2^{-Nj} \mu_j. \quad (7.3)$$

The value is therefore obtainable with finite rational arithmetic.

*Proof.* Let  $u_N$  be the density of  $\sum_{j=0}^{N-1} 2^{-j} U_j$ . Convolution of the  $N$  uniform densities gives the finite inclusion–exclusion formula

$$u_N(t) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{m=0}^{2^N-1} \sigma(m) (t + 1 - 2^{-N} - m2^{-(N-1)})_+^{N-1}. \quad (7.4)$$

To verify this, shift the uniforms to intervals  $[0, 2^{-j}]$ . The density of their sum is the derivative of the box-truncated simplex volume: inclusion–exclusion subtracts subsets of upper-bound violations, with sign  $(-1)^{|S|}$  and displacement  $\sum_{j \in S} 2^{-j}$ . These subset sums are exactly  $m2^{-(N-1)}$ , and  $|S|$  has parity  $s_2(m)$ . The reciprocal product of interval lengths is  $2^{N(N-1)/2}$ . This yields (7.4).

The remaining random tail is  $2^{-N} X'$ , independent of the first  $N$  terms. Hence  $u(x) = \mathbb{E} u_N(x - 2^{-N} X')$ . Because  $2^N x$  is even, each  $2^N a_m(x)$  is an odd integer. Thus  $|a_m(x)| \geq 2^{-N}$ . As  $|X'| \leq 1$ , the argument  $a_m(x) - 2^{-N} X'$  never changes sign except possibly at an endpoint of probability zero. The negative terms vanish; the positive terms may be expanded as ordinary powers over the whole expectation. Only the even moments survive, and the binomial expansion is precisely (7.3). At  $N = 1$  the truncated zeroth power is interpreted as the positive-half-line indicator; the same almost-sure argument applies.  $\square$

The algorithm is a transparent exact certificate, not a claim of optimal bit complexity. Its number of summands grows exponentially with the denominator depth. More efficient evaluation is a separate algorithmic question and is not needed for the proofs above.

## 7.3 Reproduced exact values and scope of testing

The bundled `verify.py` uses Python’s arbitrary-precision integers and `fractions.Fraction` for the exact part. The default run checks all eight phases  $h/8$ ,  $0 \leq h < 8$ , for meshes  $2^d$ ,  $0 \leq d \leq 4$ ; all lower-degree moments; the precise first-threshold zero pattern; and cyclic-filter reindexing examples. The arithmetic criterion for permitted frequencies is checked for reduced  $a/b$  with  $1 \leq a \leq 16$ ,  $1 \leq b \leq 12$ ,  $0 \leq r \leq 6$ , and  $1 \leq n \leq 64$ .

| $M$ | $p = v_2(M) + 1$ | $\mathcal{D}_{M,p}(0)$ | $\mathcal{D}_{M,p}(1/4)$ |
|-----|------------------|------------------------|--------------------------|
| 1   | 1                | 0                      | 13/72                    |
| 2   | 2                | 1/72                   | 0                        |
| 4   | 3                | 0                      | −4643/11059200           |
| 8   | 4                | −23/5529600            | 0                        |
| 16  | 5                | 0                      | 4966069/383551316951040  |

The four even-mesh rows reproduce values already in the repository’s comb manuscript [4]. They are cross-checks, not new numerical claims. The exact run passes 261 quadrature, moment, and reindexing assertions and 53,760 frequency-arithmetic assertions.

An optional `mpmath` part, run at 75 decimal digits, checks 240 odd-dilation ratios and their multiscale envelopes, 36 leading-zero derivative formulas, and 12 Fourier-normalization comparisons with the independent exact evaluator. The largest absolute discrepancy in the last group, using 200 positive odd harmonics, is approximately  $6.40 \times 10^{-19}$ . These are floating-point diagnostics, *not* interval enclosures. The numerical criterion used in the program is  $10^{-18}$ . No universal mathematical conclusion in this article depends on passing those finite tests.

For orientation, the first relative-tail constants are

| $p$ | $\eta_p$ (approximately) | Guaranteed range of $G_{p,q}$ |
|-----|--------------------------|-------------------------------|
| 1   | 0.2337005501             | [0.7662994499, 1.2337005501]  |
| 2   | 0.0517997903             | [0.9482002097, 1.0517997903]  |
| 3   | 0.0146780316             | [0.9853219684, 1.0146780316]  |
| 4   | 0.0045237628             | [0.9954762372, 1.0045237628]  |

The exact guaranteed endpoints are  $1 \pm \eta_p$ ; the displayed decimal endpoints are rounded illustrations, not outward-rounded certificates.

## 8 Formalization map and proof obligations

The article is independent of any unverified repository theorem because its kernel facts, Fourier criterion, and all new conclusions have explicit proofs. Nevertheless the existing Lean modules provide a natural route to formalizing the extensions. This section is a plan, not a report of a successful Lean build.

The inspected `RvachevSuperconvergentSynthesis.lean` imports `CombDefectSeries` and `RvachevPolynomialSynthesis` and exposes physical-coordinate quadrature and polynomial-deconvolution statements [3]. The canonical prose also names existing formal multiple-angle and first-defect lemmas [4]. These are useful interfaces to reuse after checking their current types at the pinned revision.

| Proposed layer            | Main formal obligations                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inverse-square comparison | Isolate the two factors at $q/2$ and $q/4$ ; prove their absolute numerator identities for odd integers; pass the finite comparison to the nonzero infinite product.          |
| Relative defect factor    | Define removable sine/cosine quotients for odd frequencies; establish the $n$ bound; use a uniform summability theorem and transfer the exact alias identity.                 |
| Signed phase filters      | Represent finite signed measures on the additive circle; justify termwise integration against finite total variation; prove Fourier uniqueness through trigonometric density. |
| Arithmetic rigidity       | Formalize the reduced-fraction condition $b \mid n$ and the subsequent 2-adic divisibility threshold.                                                                         |
| Atomic optimality         | Show that invariance permutes nonzero atoms in orbits of length $L$ and preserves the mass of each atom.                                                                      |
| Quantitative obstruction  | Form the annihilating polynomial on unit-modulus phase nodes, bound its elementary-symmetric coefficients, and compare Fourier coefficients with uniform norms.               |

Several distinctions should remain visible in a formal statement. First,  $M$  is a positive real in the analytic definition and only later specialized to an integer or reduced rational. Second,  $r$  may be zero; this is essential to the irrational obstruction. Third, phase equality is modulo one, rather than equality with just one of four literal real representatives. Fourth, support size counts distinct nonzero atoms; zero weights and duplicate listed phases must be removed before the sharp cardinality assertion. Finally, exactness is simultaneously for all  $P \in \mathcal{P}_r$  and all  $\theta$ . Dropping the latter quantifier produces a different theorem and cannot be hidden in an abbreviation.

## 9 Further research questions and topics

The following questions are not asserted to be globally open in every formulation. They are precise next tasks suggested by the proved results, and should receive their own literature checks before a novelty claim. The first is closely related to a mixed-jet nonvanishing issue retained in the repository’s comb discussion [4].

**Research question 1** (The next defect at a selected phase). For  $M = 2^d q$  and  $\theta \in Z_{d+1}$ , determine exactly when  $\mathcal{D}_{M,d+2}(\theta) = 0$ . Is the degree of precision always exactly  $d + 1$ , or do additional exceptional meshes and phases occur?

The first-defect theorem does not answer this: the next derivative involves higher jets at the odd aliases and leading jets at an additional valuation layer. A proof of noncancellation has to control both. For example, at  $M = 1$  the two selected phases fail on  $x^2$ , with defects  $-1/9$  and  $5/36$ , respectively, but finite examples do not settle the general question.

**Research question 2** (Sharp relative dominance and root conditioning). Determine the best constant  $\eta_p^*$  such that  $|G_{p,q}(\theta) - 1| \leq \eta_p^*$  for all positive odd  $q$  and all  $\theta$ . Is the supremum attained, and how does it depend on  $p$ ?

Our  $\eta_p$  comes from a termwise triangle inequality. It does not exploit the arithmetic correlation of the signs and amplitudes across distinct odd harmonics. Sharp constants could improve certified phase recovery.

**Research question 3** (Optimal quantitative under-budget error). For a rational mesh and  $K < L$ , compute or estimate sharply

$$\inf\{E_{M,r}(\nu) : \nu \text{ has mass one and at most } K \text{ atoms}\}.$$

How do positive and signed filters differ away from exactness?

Theorem 4.2 proves strict positivity without compactness assumptions on signed weights. It does not identify a minimizer or a sharp constant. Coalescing phases and unbounded signed weights are genuine issues in an optimization proof, even though they cannot drive the error to zero below the threshold.

**Research question 4** (Arithmetic behavior near rational scales). For fixed  $K$ , describe the smallness of the irrational lower bound in (4.7) as  $M$  approaches a rational mesh. Can its vanishing rate be expressed using the multiplicities of the nearby integer zeros and rational approximation of  $M$ ?

The leading-jet identity suggests a local power law near any fixed resonance. A uniform theorem over families with growing  $K$  would also need control of which resonance supplies the minimum.

*Editorial note (ProveIt, 2026-09-30).* For fixed  $K$  this question is answered by Theorem `thm:resonance` of the later unreviewed draft `Approximate_Phase_Rigidity/` of this directory (2026-09-30), for the optimal error itself and not only for the bound (4.7). Near a reduced rational  $a/b$ , the infimum of  $E_{M,r}(\nu)$  over mass-one filters with at most  $K$  atoms, positive, real signed or complex, with nodes and weights allowed to depend on  $M$ , is of exact order  $|M - a/b|^{\max\{0, v_2(a)+1+\lfloor \log_2(K/b) \rfloor - r\}}$  for  $K \geq b$ , and stays bounded away from zero for  $K < b$ . The exponent is the largest zero multiplicity of  $H$  at the frequencies  $na/b$ ,  $1 \leq n \leq K$ , minus  $r$ ; for  $r = 0$  the same multiplicity gives the vanishing order of the minimum in (4.7). An explicit regular polygon attains the order. The constants and the neighbourhood depend on  $a/b$ ,  $K$  and  $r$ ; their uniformity as  $K$  grows, and the optimal leading constants, remain open there. That draft does not compute the under-budget infimum of the preceding question at the rational mesh itself.

**Research question 5** (Several dimensions and correlated phases). For a product of Rvachev kernels, classify phase measures giving uniform exactness on rectangular or total-degree polynomial spaces. Can correlated shifts outperform the Cartesian product of one-dimensional optimal rules?

One must not simply tensor the one-dimensional necessity statement. A multidimensional alias is a product of derivative factors; a zero factor in one coordinate can annihilate the whole alias. The permitted Fourier set therefore has a different combinatorial structure, and the one-dimensional cyclic-invariance proof alone does not imply a product support lower bound. This distinction makes the correlated problem a substantive next target.

**Research question 6** (Other dilation systems). Replace the dyadic weights in the random series by  $b^{-j}$  for an integer  $b > 2$ , or by a nonintegral geometric ratio. Which phase classification, exact-filter symmetry, and finite-support rigidity statements survive?

The measure-level Fourier test survives unchanged for a smooth compactly supported kernel. The hard part is the arithmetic geometry of its zero set and multiplicities. The particularly strong two-factor half-integer comparison used here need not persist in another base.

**Research question 7** (Finite certificates and efficient formalization). Construct a certified implementation that returns either an exact cyclic symmetry certificate or a nonzero Fourier witness for a proposed filter, with rational or rigorously enclosed transcendental phase data. Formalize the main signed-filter theorem and the under-budget bound in Lean.

The finite annihilator supplies a natural witness search bound: the first  $K$  multipliers suffice below the exactness threshold. The dyadic exact evaluator supplies a separate rational route for regression tests. Neither ingredient alone proves a favorable end-to-end bit-complexity bound; that would require a specified representation of phases and controlled evaluation of the nonzero kernel coefficients.

## 10 Conclusion

Uniform phase averaging has an exact arithmetic structure. At a reduced rational mesh  $a/b$ , polynomial degree  $r$  forces invariance under a cyclic group of order  $b2^{\max(0, r-v_2(a))}$ . At the minimum support size, the only signed filter is the positive equally spaced one, and its action is exactly ordinary mesh refinement. At an irrational mesh, only continuous Haar averaging can be exact even on constants for all translations. A finite annihilating polynomial turns these structural facts into positive lower error bounds for every under-budget signed rule.

The complementary inverse-square odd-dilation estimate and its uniform log-Gaussian strengthening refine the repository’s established first-harmonic approach. It gives a single integer-mesh theorem with explicit relative constants, simple phase zeros, and a global distance-to-zero certificate, without confusing known phase classifications with new results or claiming unproved higher-order maximality. The principal remaining opportunities are sharp approximation below the phase threshold, mixed higher jets, and genuinely multidimensional phase correlations.

## References

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`d-research-frontiers/drafts/inverse-and-sampling/comb-interpolation/comb_interpolation_synthesis/chapters/03_additive_dyadic.tex`.